

Olive, soybean and palm oils intake have a similar acute detrimental effect over the endothelial function in healthy young subjects.

BACKGROUND AND AIM: Currently, more than 30% of the caloric intake in the Colombian population comes from vegetable oil consumption mainly by the ingestion of deep-fried foods. Recently, it has been reported that unsaturated fatty acid rich oils have a beneficial effect on the endothelial function. Nevertheless, it is well known that the deep-frying process alters the chemical composition of vegetable oils and can produce adverse effects in the endothelial function. **OBJECTIVE:** To evaluate the acute effect of the ingestion of large amounts of olive, soybean and palm oils, fresh and at two different deep-fry levels, on the glucose and lipid profiles and the endothelial function. **METHODS AND RESULTS:** Ten healthy young volunteers were included in the study. After performing a baseline evaluation of cardiovascular risk factors and drawing a fasting blood sample, subjects were exposed to a randomly assigned potato soup meal containing 60 mL of one of three different vegetable oils (olive, soybean and palm), either fresh or at one of two different deep-fry levels (10 and 20 fries, respectively). Flow-mediated vasodilation (FMD) was performed in fasting conditions and 3h after the intake of the oil rich meal. Furthermore, blood samples were taken at these stages for the lipid profiles and plasma glucose determinations. All the meals resulted in a similar acute endothelial impairment (FMD decrease of 32.1%, confidence interval [CI] 95%, 28.0-36.2) and postprandial increase in triglycerides (27.03%, CI 95%, 20.5-33.3), independently of the type of oil ingested ($p=0.44$) and regardless of its deep-fry level ($p=0.62$). No correlation was found between endothelial impairment and postprandial triglyceride increment ($r=-0.22$, $p=0.09$). **CONCLUSIONS:** No difference was found in the acute adverse effect of the ingestion of different vegetable oils on the endothelial function. All the vegetable oils, fresh and deep-fried, produced an increase in the triglyceride plasma levels in healthy subjects.